

VELTRAXX'26



इलेक्ट्रॉनिकी एवं
सूचना प्रौद्योगिकी मंत्रालय
MINISTRY OF
ELECTRONICS AND
INFORMATION TECHNOLOGY



Chips to Startup
Programme



INSTITUTION'S
INNOVATION
COUNCIL
(Ministry of Education Initiative)

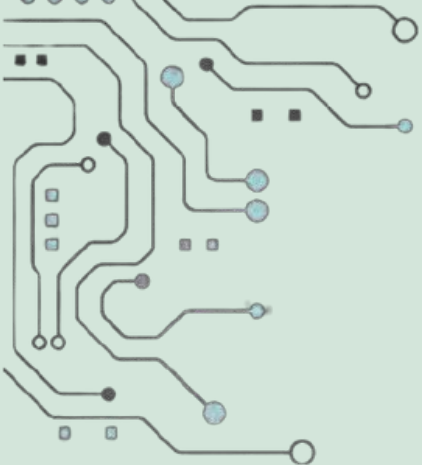
BAJAJ INSTITUTE OF TECHNOLOGY, WARDHA

TEAM NAME: NEXORA

PROBLEM STATEMENT NO: 16

PROBLEM STATEMENT TITLE: FPGA- Based Accident detection and GPS-SMS alert System

PS Category- Hardware



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PROBLEM STATEMENT

Real-world problem

- Accidents may go unreported when the driver cannot respond.
- Emergency teams may not receive the accident location quickly.

Why it matters

- Every second matters during emergency response.
- Automatic location sharing can reduce response delays.

Current challenges

- Impact does not always mean an accident.
- Potholes, braking and vibrations can cause false triggers.
- Single-sensor detection lacks reliability.



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YOUR SOLUTION

Real-time accident validation using multi-sensor fusion with automatic GPS-SMS emergency alerts.

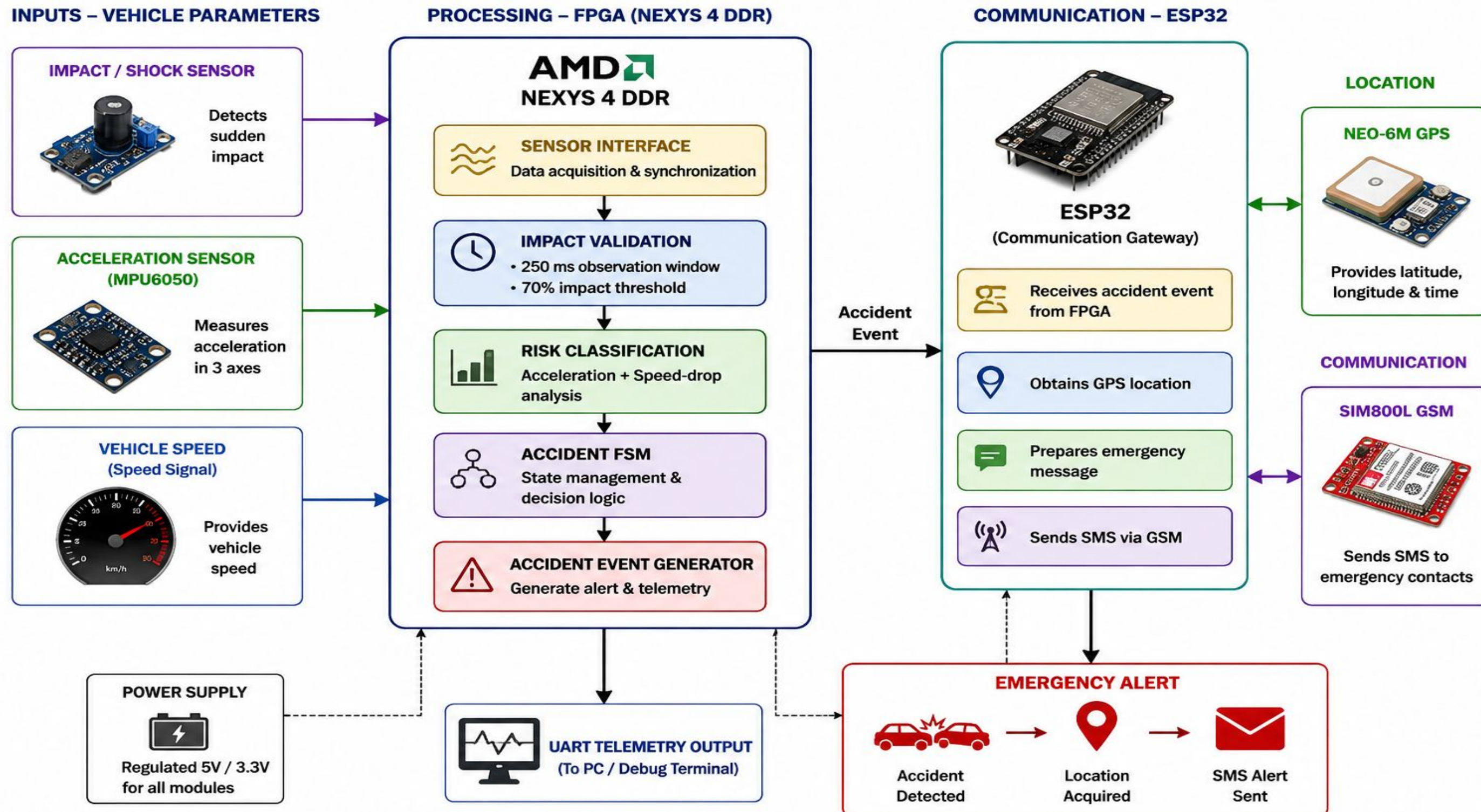
How it works

Detect → Validate → Decide → Locate → Alert

- Monitors **impact, acceleration & speed**
- **250 ms** event-validation window
- **70%** impact-validation threshold
- Acceleration + **speed-drop analysis**
- FPGA performs real-time accident decision
- ESP32 handles GPS & GSM communication

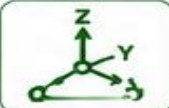
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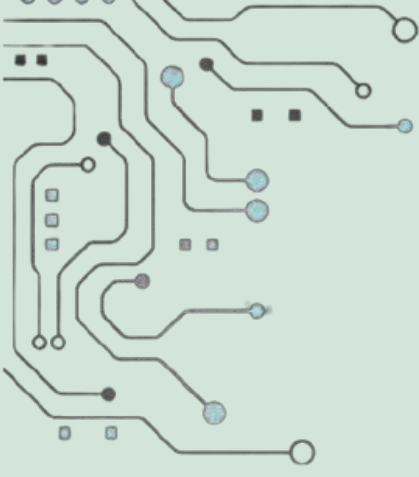
BLOCK DIAGRAM



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IDEA BEHIND EACH BLOCK & TRADE-OFFS

BLOCK	WHY WE CHOSE IT	ROLE IN NEXORA	TRADE-OFF
 Impact / Shock Sensor	Directly senses a sudden impact or collision.	First indicator of a possible accident.	 Can give false triggers due to potholes or vibrations.
 IMU / Acceleration (MPU6050)	Measures 3-axis acceleration to detect abnormal motion.	Provides acceleration data for validation and risk analysis.	 Needs calibration and filtering to avoid noise.
 Speed Input (Vehicle Speed)	Helps identify sudden speed drop with impact.	Supports accident detection through speed-drop analysis.	 Requires reliable speed source and signal conditioning.
 FPGA (Nexys 4 DDR)	Deterministic, real-time and parallel processing for critical decisions.	- Synchronizes inputs - Validates impact - Runs risk classifier & Accident FSM	 More complex design (RTL, timing, resources).
 250 ms Validation Window	Gives time to observe the event and confirm it.	Confirms that the impact is sustained within 250 ms.	 Introduces a small delay (~250 ms) in detection.
 70% Threshold	Majority of samples must indicate real impact.	If impact score $\geq 70\%$, it moves to risk classification.	 Threshold needs tuning for different vehicles & conditions.
 Accident FSM	Structured state management for stable and predictable decisions.	Manages states: Normal $\rightarrow$ Impact $\rightarrow$ Validation $\rightarrow$ Risk $\rightarrow$ Accident.	 More states and transitions increase RTL complexity.
 ESP32 (Communication Gateway)	Excellent for handling communication tasks.	- Receives event from FPGA - Gets GPS location - Sends SMS via GSM	 Additional device and firmware integration required.
 GPS (NEO-6M)	Provides latitude, longitude and time.	Provides accident location.	 Location acquisition time may vary (satellite availability).
 GSM (SIM800L)	Easy and widely used for sending SMS alerts.	Sends emergency SMS with location to contacts.	 Depends on cellular network availability.
 ARCHITECTURAL TRADE-OFF (OVERALL) We choose more reliable and deterministic accident detection with real-time decision making at the cost of increased hardware, design complexity and power.  Impact Detected $\rightarrow$  Validated Smartly $\rightarrow$  Decision Made $\rightarrow$  Location Acquired $\rightarrow$  Alert Sent			



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OUTPUTS AND RESULTS

Results

Impact Detection

- 250 ms validation
- 70% threshold evaluation

Accident Detection

- Acceleration + speed-drop analysis

FPGA Verification

- FSM verified
- Risk classifier verified
- UART telemetry verified

Simulation Result

16 / 16 checks passed

Emergency Output

GPS Location → GSM SMS Alert



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RESEARCH AND REFERENCE

FPGA Based Accident Detection and Monitoring System for Safety Traffic — Fazal Noorbasha, K. D. M. Vikash Pani, K. Seetharam, S. Narendra

Accident Detection and Reporting System Using GPS, GPRS and GSM Technology — Md. Syedul Amin, Jubayer Jalil, M. B. I. Reaz

Automated Vehicle Accident Detection and Alert System Using GSM and GPS Technology — Nagesh Mahendra Ingle, Swapnil Bhika Billore, Yogesh Kumar, Shashank Kumar Singh, Anirban J. Hati

IoT Based Automatic Vehicle Accident Alert System — Nazia Parveen, Ashif Ali, Aleem Ali